

Problem 1.1

Find all solutions z of the following equations:

- (a) $z^2 = -15 - 8i$.
(b) $(3 + 5i)z^2 + (3 + 5i)z + (8 + 2i) = 0$.

Solution. (a):

Problem 1.2

Sketch the following sets:

- (a)
 $\left\{ z \in \mathbb{C} : \frac{|z - 3i|}{|z + 3i|} \leq 2 \right\},$
(b)
 $\left\{ z \in \mathbb{C} : \operatorname{Im}\left(\frac{z-1}{1+i}\right) \geq 0 \right\},$
(c)
 $\left\{ z \in \mathbb{C} : \frac{|z-1|}{|z+2i|} < \sqrt{2} \right\}.$

Solution.

Problem 1.3

Find $f(M)$ for the following maps $f : \mathbb{C} \rightarrow \mathbb{C}$ and the following sets $M \subseteq \mathbb{C}$:

- (a) $f(z) = e^z$ and $M = \{z \in \mathbb{C} : 2 \leq \operatorname{Re}(z) \leq 3 \text{ and } |\operatorname{Im}(z)| \leq \frac{\pi}{2}\}.$
(b) $f(z) = z^2$ and $M = \{z \in \mathbb{C} : \operatorname{Im}(z) > 0 \text{ and } 1 \leq |z| \leq 2\}.$

Solution. (a): Let $z = a + bi$ for $a, b \in \mathbb{R}$. Then

$$\begin{aligned} f(z) &= e^z \\ &= e^{a+bi} \\ &= e^a e^{bi}. \end{aligned}$$

Thus

$$f(M) := \left\{ re^{i\theta} : 2 \leq r \leq 3, |\theta| \leq \frac{\pi}{2} \right\}$$

(b):

$$M = \mathbb{H} \cap \{re^{i\theta} : 1 \leq r \leq 2, \theta \in [0, 2\pi)\} = \{re^{i\theta} : 1 \leq r \leq 2, \theta \in (0, \pi)\}$$

Thus

$$\begin{aligned} f(M) &= \{r^2 e^{i2\theta} : 1 \leq r \leq 2, \theta \in (0, \pi)\} \\ &= \{re^{i\theta} : 1 \leq r \leq 4, \theta \in (0, 2\pi)\} \end{aligned}$$

Problem 1.4

- (a) Show that $(\cos x + i \sin x)^n = \cos(nx) + i \sin(nx)$ for all $n \in \mathbb{N}$, $x \in \mathbb{R}$.
 (b) Express $\cos(5x)$ in terms of $\cos x$, and $\sin(5x)$ in terms of $\sin x$.
 (c) Express $\cos(2\pi/5)$ and $\sin(2\pi/5)$ in terms of roots of real numbers.

Solution. (a): Follows immediately from Euler's Identity.

(b):

$$\begin{aligned} e^{i5x} &= \sum_{k=0}^5 \binom{5}{k} i^k \cos^{5-k}(x) \sin^k(x) \\ &= \cos^5(x) + 5i \sin(x) \cos^4(x) - 10 \sin^2(x) \cos^3(x) - 10i \sin^3(x) \cos^2(x) + 5 \sin^4(x) \cos(x) + i \sin^5(x). \end{aligned}$$

Hence:

$$\begin{aligned} \cos(5x) &= \Re(e^{i5x}) \\ &= \cos^5(x) - 10 \sin^2(x) \cos^3(x) + 5 \sin^4(x) \cos(x) \\ &= 16 \cos^5(x) - 20 \cos^3(x) + 5 \cos(x) \\ \sin(5x) &= 16 \sin^5(x) - 20 \sin^3(x) + 5 \sin(x) \end{aligned}$$

(c): Let $x = \cos(2\pi/5)$ and $y = \sin(2\pi/5)$.

$$\begin{aligned} 1 &= 16x^5 - 20x^3 + 5x \\ 0 &= 16y^5 - 20y^3 + 5y = y(16y^4 - 20y^2 + 5) \end{aligned}$$

Thus

$$\begin{aligned}
y^4 - \frac{5}{4}y^2 &= -\frac{5}{16} \\
y^4 - \frac{5}{4}y^2 + \frac{25}{64} &= \frac{5}{64} \\
\left(y^2 - \frac{5}{8}\right)^2 &= \frac{5}{64} \\
y^2 - \frac{5}{8} &= \frac{\sqrt{5}}{8} \\
y^2 &= \frac{5 + \sqrt{5}}{8} \\
y &= \frac{\sqrt{5 + \sqrt{5}}}{2\sqrt{2}}
\end{aligned}$$

Using the Pythagorean identity, we can get x .

$$\begin{aligned}
x &= \sqrt{1 - y^2} \\
x &= \frac{\sqrt{3 + \sqrt{5}}}{2\sqrt{2}}
\end{aligned}$$

Problem 1.5

Suppose that $z, w \in \mathbb{C}$ and $|z|, |w| < 1$. Show that then

$$\left| \frac{z - w}{1 - \bar{z}w} \right| < 1.$$

Solution.

$$\begin{aligned}
1 - \left(\frac{z - w}{1 - \bar{z}w} \right) \left(\frac{\bar{z} - \bar{w}}{1 - z\bar{w}} \right) &= 1 - \frac{|z|^2 - w\bar{z} - \bar{z}w + |w|^2}{1 - z\bar{w} - \bar{z}w + |zw|^2} \\
&= \frac{1 + |zw|^2 - |z|^2 - |w|^2}{|1 - \bar{z}w|^2} \\
&= \frac{(1 - |z|^2)(1 - |w|^2)}{|1 - \bar{z}w|^2} > 0
\end{aligned}$$

Problem 1.6

Let $P(z) = a_0 + a_1z + \dots + a_{n-1}z^{n-1} + z^n$ be a polynomial of degree $n \in \mathbb{N}$ with real coefficients $a_0, \dots, a_{n-1} \in [0, 1]$.

Show that if $P(z_0) = 0$ for some $z_0 \in \mathbb{C}$, then $\operatorname{Re}(z_0) < 0$ or $|z_0| < \frac{1+\sqrt{5}}{2}$.

Solution. Note that if z_0 is a solution, $\bar{z}_0$ is a solution to P as well. Thus WLOG $\arg(z_0) \in [0, \pi]$. Assume the contrary that $|z_0| = r \geq \frac{1+\sqrt{5}}{2}$ and $\arg(z) \in [0, \frac{\pi}{2}]$. We know that

$$\begin{aligned}
0 = P(z_0) &= \sum_{k=0}^{n-1} a_k z_0^k + z_0^n \\
-z_0^n &= \sum_{k=0}^{n-1} a_k z_0^k \\
r^n e^{in\theta+i\pi} &= \sum_{k=0}^{n-1} a_k r^k e^{ik\theta} \\
r^n &= \sum_{k=0}^{n-1} a_k r^k e^{i(\theta(k-n)-\pi)} \\
&= \Re \left(\sum_{k=0}^{n-1} a_k r^k e^{i(\theta(k-n)-\pi)} \right) \\
&\leq \sum_{k=0}^{n-1} r^k \max(0, \Re(e^{i(\theta(k-n)-\pi)})) \\
&= \sum_{k=0}^{n-1} r^k \max(0, \cos(\theta(k-n) - \pi)).
\end{aligned}$$

Thus we have the following inequalities:

$$\left(\frac{1 + \sqrt{5}}{2} \right)^n \leq r^n \leq \sum_{k=0}^{n-1} r^k \max(0, -\cos(\theta(n-k))).$$

Note we are using the trigonometric identity, $\cos(\alpha - \pi) = -\cos(\alpha)$. Let $k = n - 1$,

$$\cos(\theta(n-k)) = \cos(\theta) > 0 \Rightarrow \max(0, -\cos(\theta(n-k))) = 0.$$

Thus

$$\begin{aligned}
r^n &\leq \sum_{k=0}^{n-2} r^k \max(0, -\cos(\theta(n-k))) \\
&\leq \sum_{k=0}^{n-2} r^k \\
&= \frac{1 - r^{n-1}}{1 - r} = \frac{r^{n-1} - 1}{r - 1} \\
r \geq \frac{1 + \sqrt{5}}{2} &\Rightarrow r^2 - r - 1 \geq 0 \Rightarrow r^2 - r \geq 1 \Rightarrow r - 1 \geq \frac{1}{r}
\end{aligned}$$

Thus

$$r^n \leq \frac{r^{n-1} - 1}{\frac{1}{r}} \Rightarrow r^{n-1} \leq r^{n-1} - 1$$

which is a contradiction.

Problem 1.7

(a) Let $P(z) = a_0 + a_1z + \dots + a_{n-1}z^{n-1} + z^n$ be a polynomial of degree $n \in \mathbb{N}$ with complex coefficients $a_0, \dots, a_{n-1}$. Suppose that $P(z) = \prod_{j=1}^n (z - z_j)$. Find an expression for $\sum_{j=1}^n z_j^2$ in terms of the coefficients of P .

(b) Let $n \in \mathbb{N}$ and $\zeta_k = \exp\left(\frac{2\pi i k}{n}\right)$ for $k = 1, \dots, n-1$. Find a simple expression for

$$\sum_{k=1}^{n-1} \frac{1}{(\zeta_k - 1)^2}$$

in terms of n .

Solution. (a):

$$\begin{aligned} \sum_{j=1}^n z_j^2 &= \left(\sum_{j=1}^n z_j \right)^2 - \sum_{i=1}^n \sum_{j=1}^n z_i z_j \\ &= a_{n-1}^2 - 2a_{n-2} \end{aligned}$$

(b): Examine the following polynomials:

$$\begin{aligned} P(x) &= \prod_{k=1}^{n-1} (\zeta_k - x) = 1 + x^2 + \dots + x^{n-1} \\ P'(x) &= \sum_{k=1}^{n-1} \prod_{j \neq k} (\zeta_j - x) = 1 + 2x + 3x^2 + \dots + (n-1)x^{n-2} \\ P''(x) &= 2 + 6x + \dots + (n-1)(n-2)x^{n-3} = \sum_{k=1}^{n-2} k(k+1)x^{k-1}. \end{aligned}$$

Note that

$$\frac{P'(x)}{P(x)} = \sum_{k=1}^{n-1} \frac{1}{(\zeta_k - x)} \Rightarrow \frac{d}{dx} \left(\frac{P'(x)}{P(x)} \right) = \sum_{k=1}^{n-1} \frac{1}{(\zeta_k - x)^2}.$$

Thus

$$S = \sum_{k=1}^{n-1} \frac{1}{(\zeta_k - 1)^2} = \frac{P''(1)P(1) - (P'(1))^2}{(P(1))^2}$$

Now

$$\begin{aligned}
P(1) &= n \\
P'(1) &= \frac{n(n-1)}{2} \\
P''(1) &= \sum_{k=1}^{n-2} (k^2 + k) \\
&= \sum_{k=1}^{n-2} k^2 + \frac{(n-2)(n-1)}{2} \\
&= \frac{(n-2)(n-1)(2n-1)}{6} + \frac{(n-2)(n-1)}{2}
\end{aligned}$$

Hence

$$S = \frac{1}{n^2} \left(\frac{n(n-2)(n-1)(2n-1) + 3n(n-2)(n-1)}{6} - \frac{n^2(n-1)^2}{4} \right)$$

Problem 1.8

Let $n \in \mathbb{N}$. Find all (complex) roots of the polynomial

$$P(z) = (z+1)^n - z^n$$

and express them explicitly using trigonometric functions (applied to real arguments).

Solution. Suppose z is a solution to the polynomial described above. Then we have that

$$(z+1)^n = z^n \Rightarrow |z+1|^n = |z|^n \Rightarrow |z+1| = |z|.$$

Thus we have

$$\begin{aligned}
|z|^2 &= (z+1)(\bar{z}+1) \\
&= |z|^2 + 2\Re(z) + 1.
\end{aligned}$$

This implies $\Re(z) = -\frac{1}{2}$. Thus $y = \Im(z) = \frac{1}{i}(z + \frac{1}{2})$. Thus we have that

$$0 = P\left(-\frac{1}{2} + yi\right) = \left(iy + \frac{1}{2}\right)^n - \left(iy - \frac{1}{2}\right)^n$$

Meaning we want y such that

$$\left(\frac{1}{2} + iy\right)^n = \left(-\frac{1}{2} + iy\right)^n.$$

Let $\theta = \arctan(2y)$ and $r = \sqrt{\frac{1}{4} + y^2}$. Thus

$$(re^{i\theta})^n = (-re^{-i\theta})^n \Rightarrow e^{in\theta} = (-1)^n e^{-in(\theta)}$$

Thus we have 2 cases n even and n odd.

1. Case 1 n even: We have that $\sin(n\theta) = 0$, so $n\theta = k\pi$ for some $k \in \mathbb{Z}$, i.e. $\theta = \frac{k\pi}{n}$.
2. Case 2 n odd: We have $\cos(n\theta) = 0$, so $n\theta = \frac{\pi}{2} + k\pi$ for some $k \in \mathbb{Z}$, i.e. $\theta = \frac{(2k+1)\pi}{2n}$.

Since $\theta = \arctan(2y)$ ranges over $(-\frac{\pi}{2}, \frac{\pi}{2})$, only the integers k for which the corresponding θ lies in this interval give valid solutions; in each case this selects exactly $n-1$ values of k , consistent with P having degree $n-1$ (the z^n terms cancel).

Recovering y from $\tan(\theta) = 2y$ and then $z = -\frac{1}{2} + iy$ gives

$$z = -\frac{1}{2}(1 - i \tan(\theta)).$$

Explicitly, the roots are

$$z_k = -\frac{1}{2} \left(1 - i \tan\left(\frac{k\pi}{n}\right) \right), \quad -\frac{n}{2} < k < \frac{n}{2} \quad (n \text{ even}),$$

$$z_k = -\frac{1}{2} \left(1 - i \tan\left(\frac{(2k+1)\pi}{2n}\right) \right), \quad -\frac{n+1}{2} < k < \frac{n-1}{2} \quad (n \text{ odd}).$$

Problem 1.9

(a) Show that a map $L : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ is linear (as a map on the vector space $\mathbb{R}^2$ over $\mathbb{R}$) if and only if there are constants $\alpha, \beta \in \mathbb{C}$ such that L can be written in the form

$$L(z) = \alpha z + \beta \bar{z}, \quad z \in \mathbb{C},$$

under the identification of points $w \in \mathbb{C}$ with $(\operatorname{Re}(w), \operatorname{Im}(w)) \in \mathbb{R}^2$.

Hint: The map L can be represented by a matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ with respect to the standard basis in $\mathbb{R}^2$. Express α and β in terms of the matrix coefficients a, b, c, d and vice versa.

(b) When is L a $\mathbb{C}$ -linear map, that is, linear as a map on the vector space $\mathbb{C}$ over $\mathbb{C}$? Find necessary and sufficient conditions in terms of α, β and in terms of a, b, c, d .

(c) Express $\det(L)$ and the operator norm

$$\|L\| := \sup\{|L(z)| : z \in \mathbb{C}, |z| \leq 1\}$$

in terms of α and β .

Solution. $\Rightarrow$: Let $\alpha = \frac{1}{2}(a-d) + \frac{i}{2}(c+b)$ and $\beta = \frac{1}{2}(a+d) + \frac{i}{2}(c-b)$ given the matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$. Suppose $\alpha = \alpha_1 + i\alpha_2$ and $\beta = \beta_1 + i\beta_2$. Now you just bash to get a, b, c, d in terms of α_i, β_j .

$\Leftarrow$:

$$L(mz + nw) = \alpha(mz + nw) + \beta(\overline{mz + nw}) =$$

Problem 2.1

Let (X, d) be a metric space. Show that if $A, B \subseteq X$, then $\overline{A \cup B} = \overline{A} \cup \overline{B}$.

Solution. Suppose $x \in \overline{A}$, then $x \in A$ or $x \in \partial A$, in either case $x \in \overline{A \cup B}$. By a symmetric argument, the same is true for $x \in \overline{B}$. Thus $\overline{A} \cup \overline{B} \subseteq \overline{A \cup B}$. Suppose $x \in \overline{A \cup B}$, $x \in A \cup B$ or $x \in \partial(A \cup B)$. The second case, is the one which is remotely interesting. x is the limit point of a sequence $\{x_n\}_{n=0}^{\infty}$ without loss of generality $x_n \in A$ (otherwise in B or pass to a subsequence). Thus $x \in \overline{A} \subseteq \overline{A \cup B}$. Thus $\overline{A \cup B} = \overline{A} \cup \overline{B}$.

Problem 2.2

Let (X, d) be a metric space and $M \subseteq X$. Recall that a set $A \subseteq M$ is called *relatively closed* (in M) if there exists a closed set $A' \subseteq X$ such that $A = A' \cap M$.

- (a) Show that $A \subseteq M$ is relatively closed if and only if $A = \overline{A} \cap M$.
 (b) Show that $A \subseteq M$ is relatively closed if and only if the following condition is true: whenever $\{x_n\}$ is a convergent sequence in A and $x_n \rightarrow x \in M$, then $x \in A$.

Solution. (a): Suppose $A \subseteq M$ is relatively closed. Then $\exists B' \subseteq X$ closed such that $A = B' \cap M$. Note that, $A \subseteq \overline{A} \subseteq B'$. Thus we have that $A \subseteq \overline{A} \cap M \subseteq B' \cap M = A \Rightarrow \overline{A} \cap M = A$. The other direction follows immediately from the definition of relatively closed.

(b): For the forward direction apply a. Suppose for all convergent sequences $\{x_n\} \in X$ where $x_n \rightarrow x \in M$, then $x \in A$. This kind of immediately implies the $\overline{A} \cap M \subseteq A$ which implies equality.

Problem 2.3

- (a) Show that if $A \subseteq \mathbb{C}$ is a closed set contained in an open disk $D = B(z_0, R)$, where $z_0 \in \mathbb{C}$ and $R > 0$, then there exists a number $r < R$ such that $|z - z_0| \leq r$ for all $z \in A$.
 (b) For non-empty sets A and B in $\mathbb{C}$ we define their distance as

$$\text{dist}(A, B) := \inf\{|z - w| : z \in A \text{ and } w \in B\}.$$

Show that if A is compact and B is closed, then there exist points $u \in A$ and $v \in B$ such that $\text{dist}(A, B) = |u - v|$.

Solution. (a): Consider the set $A_D = \{|z - z_0| : z \in A\}$. Note that $A \subseteq D$, thus $|z - z_0| < R$. Thus A_D is bounded above by R . Thus $\exists r \leq R$ such that $\sup A_D = r$. We now need to show that $r < R$. Assume the contrary, that $r = R$. Thus for all ε , $\exists z \in A$ such that $|z - z_0| \in (R - \varepsilon, R)$. Now consider the sequence $(x_n)_{n=1}^{\infty}$ where $x_n \in A$ and $|z_0 - x_n| \in (R - \frac{1}{n}, R)$. By Bolzano Weierstrass, x_n has a convergent subsequence, without loss of generality pass to it. Let x^* be the limit point. Note that the function $x \rightarrow |z_0 - x|$ is continuous hence $|z_0 - x^*| = R$ but $x^* \in A \subset D$ and a contradiction arises. Thus $r < R$.

(b): Suppose A is compact and B is closed. Consider

$$d(x, B) := \inf\{|z - x| : z \in B\}.$$

$d(\cdot, B) : \mathbb{C} \rightarrow \mathbb{R}$ there exists a $z \in B$ such that $|z - x| = d(x, B)$, this is by intersecting B with a closed ball and then compactness applies (Heine Borel) and then minimum is achieved. Now we have to show that $f = d(\cdot, B)$ is a continuous function. $\forall \varepsilon > 0$, $\forall x \in f^{-1}([0, \varepsilon))$ we

know that $f(x) < \varepsilon$. We want to show there exists a $B(x, \delta) \subseteq f^{-1}([0, \varepsilon))$. We know that $\exists b \in B$ such that $|x - b| = f(x) < \varepsilon$. $\forall y \in B(x, \varepsilon - |x - b|)$ we have:

$$f(y) \leq |y - b| \leq |x - y| + |x - b| < \varepsilon - |x - b| + |x - b| = \varepsilon \Rightarrow y \in f^{-1}([0, \varepsilon))$$

We will also show that $f^{-1}((\varepsilon, \infty))$ is open. $\forall x \in f^{-1}((\varepsilon, \infty))$, $f(x) > \varepsilon$. Assume the contrary, $\forall n \in \mathbb{N}$ there exists $y_n \in B(x, \frac{1}{n})$ such that $f(y_n) < \varepsilon$ which means there are $b_n \in B$ such that $|y_n - b_n| < \varepsilon$. Thus

$$|x - b_n| \leq |x - y_n| + |y_n - b_n| < \frac{1}{n} + \varepsilon \Rightarrow f(x) \leq \varepsilon$$

thus a contradiction occurs. Hence $\exists n \in \mathbb{N}$ such that $B(x, \frac{1}{n}) \subset f^{-1}((\varepsilon, \infty))$. Thus f continuous. Now $\exists a \in A$ such that $f(a) \leq f(a')$ for all $a' \in A$. This effectively solves the problem.

Problem 2.4

- (a) Show that the function $z \mapsto \frac{z^2+3}{z+1}$ is continuous on the set $\mathbb{C} \setminus \{-1\}$ by using the ε - δ -definition of continuity.
- (b) Suppose $\{z_n\}$ and $\{w_n\}$ are convergent sequences in $\mathbb{C}$, say $z_n \rightarrow z$ and $w_n \rightarrow w$. Assume $w_n \neq 0$ for $n \in \mathbb{N}$ and $w \neq 0$. Show that $z_n/w_n \rightarrow z/w$ by using the ε -definition for convergence.

Solution. (a): The product of continuous $f : \Omega_1 \rightarrow \mathbb{C}$ and $g : \Omega_2 \rightarrow \mathbb{C}$ is continuous on $\Omega_1 \cap \Omega_2$ (Ω_1 and Ω_2 are open). $\forall x \in \Omega_1 \cap \Omega_2$, there exists a $\delta_0 > 0$ such that $B(x, \delta_0) \subseteq \Omega_1 \cap \Omega_2$. Thus $\overline{B}(x, \frac{\delta_0}{2}) \subset \Omega_1 \cap \Omega_2$. Thus f and g achieve a maximum on $\overline{B}(x, \frac{\delta_0}{2})$ denoted by M_f and M_g . Now $\forall \varepsilon > 0$ there exist a $\delta < \frac{\delta_0}{2}$ such that $|f(x) - f(y)| < \frac{\varepsilon}{2M_g}$ and $|g(x) - g(y)| < \frac{\varepsilon}{2M_f}$ for $y \in B(x, \delta)$.

$$\begin{aligned} |f(x)g(x) - f(y)g(y)| &\leq |f(x)g(x) - f(x)g(y)| + |f(x)g(y) - f(y)g(y)| \\ &\leq M_f|g(x) - g(y)| + M_g|f(x) - f(y)| < \varepsilon \end{aligned}$$

Let $f(x) = x^2 + 3$ and $g(x) = (x + 1)^{-1}$. f and g are continuous on $\mathbb{C}$ and $\mathbb{C}/\{-1\}$ respectively. Thus the function given is continuous on $\mathbb{C}/\{-1\}$.

(b):

$$\begin{aligned} \left| \frac{z_n}{w_n} - \frac{z}{w} \right| &= \left| \frac{z_n w - z w_n}{w_n w} \right| \\ &= \left| \frac{z_n w - z w + z w - z w_n}{w_n w} \right| \\ &\leq \left| \frac{w(z_n - z)}{w_n w} \right| + \left| \frac{z(w - w_n)}{w_n w} \right| \\ &\leq \frac{1}{m_w} |z_n - z| + \frac{1}{m_w} \left| \frac{z}{w} \right| |w - w_n| \end{aligned}$$

Problem 2.5

Suppose $f : \overline{\mathbb{D}} \rightarrow \mathbb{C}$ is a continuous function with $\exp(f(z)) = 1$ for all $z \in \overline{\mathbb{D}}$. Show that then f is constant.

Solution. Note that the range of f must be $\{2\pi i k : k \in \mathbb{N}\}$ which is discrete. The only continuous function from a euclidean space to discrete space is constant.

Problem 2.6

Let $U \subseteq \mathbb{C}$ be an open set with $0 \in U$. Show that then it is impossible to define a continuous square root function on U ; more precisely, show that there exists no continuous function $S : U \rightarrow \mathbb{C}$ such that $S(z)^2 = z$ for each $z \in U$.

Hint: Consider a suitable parametrization of a small circle centered at 0.

Solution. There exists a $r > 0$ such that $\overline{B}(0, r) \subset U$. Assume for the sake of contradiction, that there exists a continuous function $S : U \rightarrow \mathbb{C}$ such that $S(z)^2 = z$. We will examine the behavior close to $-r \in U$ specifically along the paths $\theta \rightarrow e^{ri\theta}$ and $\theta \rightarrow e^{-ri\theta}$ as $\theta \in [0, \pi]$. Let $\gamma : [-\pi, \pi] \rightarrow \mathbb{C}$ where $\theta \rightarrow e^{i\theta}$. γ is continuous and hence $S \circ \gamma$ is continuous. $S(\gamma(\theta)) = c(\theta)\sqrt{r}e^{i\frac{\theta}{2}}$ where $c(\theta) \in \{-1, 1\}$ and continuous. Thus $c(\theta) = C \in \{-1, 1\}$. As $\theta \rightarrow -\pi$, $S(\gamma(\theta)) \rightarrow C\sqrt{r}e^{-i\frac{\pi}{2}} = -Ci\sqrt{r}$ while $\theta \rightarrow \pi$, $S(\gamma(\theta)) \rightarrow Ci\sqrt{r}$. Thus $S(-r)$ has no consistent value and thus S cannot be continuous.

Problem 2.7

Let (X, d) be a metric space. Recall that $M \subseteq X$ is called *connected* if the following condition is true: if $U, V \subseteq X$ are open, $M \subseteq U \cup V$, and $U \cap V = \emptyset$, then $M \subseteq U$ or $M \subseteq V$.

Show that $M \subseteq X$ is connected if and only if $\emptyset$ and M are the only subsets of M that are both relatively open and relatively closed.

Note: This statement would be very easy to prove if we replaced the condition $U \cap V = \emptyset$ with $U \cap V \cap M = \emptyset$ in the definition of connectedness.

Solution. $\Rightarrow$: Suppose $M \subseteq X$ is connected. Suppose $\emptyset \neq S \subseteq M$ is relatively open and relatively closed. There exists $A \subset X$ open such that $A \cap M = S$ and similarly $B \subset X$ closed such that $B \cap M = S$.

$$\begin{aligned} M/S &= M \cap (A \cap M)^c = M \cap (A^c \cup M^c) = M \cap A^c \\ M/S &= M \cap (B \cap M)^c = M \cap B^c, \end{aligned}$$

Thus S and M/S are both relatively clopen in M . Consider $\forall x \in S$, there exists a $\varepsilon_x > 0$ such that $B(x, \varepsilon_x) \subset A \Rightarrow B(x, \varepsilon_x) \cap M/S = \emptyset$. Then $d(x, M/S) \geq \varepsilon_x$. Similarly $\forall x \in M/S$ there exists a $\delta_x > 0$ such that $d(x, M) > \delta_x$. Consider

$$U = \bigcup_{x \in S} B\left(x, \frac{d(x, M/S)}{4}\right)$$

$$V = \bigcup_{x \in M/S} B\left(x, \frac{d(x, S)}{4}\right).$$

Assume the contrary that $\exists x \in U \cap V$, thus $\exists s \in S$ and $t \in M/S$ such that $d(x, s) < d(s, M/S)$ and $d(x, t) < d(t, S)$. Thus

$$\begin{aligned} d(s, t) &\leq d(x, s) + d(x, t) \\ &< \frac{d(s, M/S)}{4} + \frac{d(t, S)}{4} < \frac{1}{2}d(s, t) \end{aligned}$$

This is a contradiction, hence $U \cap V = \emptyset$. Moreover $M = S \cup (M/S) \subset U \cup V$. Neither S and M/S are empty thus M cannot be connected.

$\Leftarrow$: Suppose $\emptyset, M$ are the only relatively clopen sets in M . Assume the contrary, that $\exists U, V \subset X$ such that $M \subseteq U \cup V$ but their intersections are empty and $M \cap U \subset U$ and $M \cap V \subset M$. Then $U \subseteq V^c$,

$$M \cap V^c \subseteq (U \cup V) \cap V^c \subseteq U \cap V^c = U \Rightarrow M \cap V^c = M \cap U.$$

Thus $M \cap U = M \cap V^c$ is relatively clopen and not all of M but isn't empty either.

Problem 2.8

Let (X, d) be a metric space, and $A \subseteq X$ be a connected set. Show that every set $B \subseteq X$ with $A \subseteq B \subseteq \overline{A}$ is also connected.

Solution. Connectedness definition is equivalent to the same definition replacing U and V with closed sets just take U^c and V^c . Assume for the sake of taking the contrapositive that B is “between A and $\overline{A}$ ” and is not connected thus $B \subset U \cup V$ where U and V are closed and $U \cap V \neq \emptyset$ but $B \not\subset U$ and $B \not\subset V$. Note that U and V form a disconnection for $\overline{A}$ as well. Furthermore, if $A \subset U$ or $A \subset V$ then so is $\overline{A}$ thus U and V form a disconnection of A as well and thus A cannot be connected. By the contrapositive the statement is true.

Problem 2.9

Let (X, d) be a metric space, I be an index set, and $M_i \subseteq X$ be connected for $i \in I$. Show that if $M = \bigcap_{i \in I} M_i \neq \emptyset$, then $\bigcup_{i \in I} M_i$ is connected.

Solution. We will prove this using the contrapositive. Suppose that the M is not connected. Thus there exists disjoint open sets U and V such that M is in $U \cup V$ but $M \not\subset U$ and $M \not\subset V$.

$$J = \{i \in I : M_i \cap U \neq \emptyset\}$$

$$K = \{i \in I : M_i \cap V \neq \emptyset\}$$

Assume the contrary, that there exists $i \in J \cap K$, thus $M_i \subset U \cup V$ and $M_i \cap U \neq \emptyset$ and $M_i \cap V \neq \emptyset$. Since U and V are disjoint, they are a disconnection of M_i and hence M_i is not connected which is a contradiction. Thus

$$\begin{aligned} \bigcup_{i \in J} M_i &= U \\ \bigcup_{k \in K} M_k &= V \\ \bigcup_{i \in J} M_i \cap V &= \emptyset \\ \bigcap_{i \in I} M_i &\subseteq \left(\bigcup_{i \in J} M_i \right) \cap \left(\bigcup_{k \in K} M_k \right) \subseteq \left(\bigcup_{i \in J} M_i \right) \cap V = \emptyset. \end{aligned}$$

Thus by the contrapositive, the statement is true.

Problem 2.10

Let M be a non-empty subset of $\mathbb{C}$. A (*connected*) *component* C of M is a maximal (with respect to inclusion) connected subset of M . Show that:

- (a) Each component C of M is relatively closed in M .
- (b) If C and C' are distinct components of M , then $C \cap C' = \emptyset$.
- (c) Every connected subset A of M lies in a unique component C of M .
- (d) M is the (disjoint) union of its components.
- (e) There exists an equivalence relation $\sim$ on M whose equivalence classes are the components of M . Give a concise description of $\sim$ that does not use the concept of a component!

Solution. **(a):** Since C is connected in M , $\overline{C} \cap M$ is also connected in M . But since C is maximal, $\overline{C} \cap M = C \Rightarrow C$ relatively closed.

(b): Suppose C and D are distinct components of M . Use 2.9 to contradict.

(c): Combine maximality and claim b.

(d): Each singleton of M is a connected subset, each of which is in one unique component.

(e): The components define a partition of M . $x \sim y \Leftrightarrow \exists \{U_i\}_{i=1}^n \subseteq \mathcal{T}(M)$ such that $x \in U_1$ and $y \in U_n$ and $U_i \cap U_{i+1} \neq \emptyset$